

Ideation Phase

Define Problem Statement

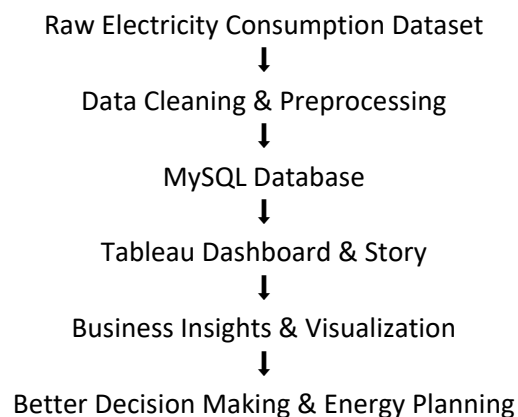
Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Project: Electricity Consumption Analysis

PS-1
I am: Government Energy Officer
I'm trying to: Monitor electricity consumption across different states and regions, identify high-demand areas, and analyze monthly electricity usage trends.
But: The electricity data is distributed across multiple records and is difficult to analyze manually.
Because: There is no centralized dashboard that provides state-wise, region-wise, and monthly electricity consumption insights.
Which makes me feel: Frustrated because decision-making becomes slower, energy planning becomes difficult, and resource allocation is less efficient.

PS-2
I am: Energy Data Analyst
I'm trying to: Analyze electricity consumption patterns using interactive dashboards and identify trends that support better planning.
But: The raw dataset contains thousands of records that are difficult to interpret without proper visualization tools.
Because: Manual analysis consumes time and important consumption trends may be overlooked.
Which makes me feel: Inefficient because generating reports and business insights becomes time-consuming.

Problem to Solution Flow



Summary of the Problem

Electricity consumption data generated across different states and regions is difficult to analyze manually due to its large volume and lack of interactive visualization tools. Government departments, electricity boards, and business analysts need an integrated solution that enables efficient analysis, state-wise and region-wise comparison, monthly trend identification, and business insights. The proposed **Electricity Consumption Analysis** system addresses these challenges using **MySQL**, **Tableau Desktop**, **Tableau Public**, and **Flask**, providing an interactive dashboard that supports faster, data-driven decision-making.